

# Speed of Light, Gravity, and Mass from Geometry

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## 1. Speed of Light (m/s)

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**Core (integers only):**

$$c_{\text{core}} = 144^3 \times 100 = 298,598,400 \text{ m/s}, \quad c_{\text{SI}} = 299,792,458 \text{ m/s}, \quad \Delta = 0.398\%$$

**Koch (geometric):**

$$c_{\text{Koch}} = (\phi^6 \times 31)^3 \times \sqrt{3} \times \left(1 + \frac{4}{720}\right) = 299,798,395 \text{ m/s}, \quad \Delta \approx 0.002\% \text{ (5,937 m/s)}$$

## 2. Gravitational Constant ( $\text{m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$ )

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**Core:**

$$G_{\text{core}} = \frac{31^2}{144} \times 10^{-11} \approx 6.673\,611\,111 \times 10^{-11}$$

$$\Delta_{\text{NIST}'26} = -0.000\,259 \times 10^{-11} \text{ (}-0.0039\%), \quad \Delta_{\text{CODATA}'22} = -0.000\,689 \times 10^{-11} \text{ (}-0.0103\%)$$

**Koch (geometric):**

$$G_{\text{Koch}} = \frac{31^2 \times \sqrt{5}}{\phi^{12} \times 10^{11}} \approx 6.673\,546\,744\,96 \times 10^{-11}$$

$$\Delta_{\text{NIST}'26} = -0.000\,323 \times 10^{-11} \text{ (}-0.00484\%), \quad \Delta_{\text{CODATA}'22} = -0.000\,753 \times 10^{-11} \text{ (}-0.01129\%)$$

## 3. Proton Mass ( $\text{MeV}/c^2$ )

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$$m_p = 31^2 - \frac{1000}{144 - 100} = 961 - \frac{1000}{44} \approx 938.272\,727 \dots$$

$$\text{CODATA: } 938.272\,081\,3, \quad \Delta = 0.000\,069\%$$

## 4. Neutron-to-Proton Mass Ratio

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$$\frac{m_n}{m_p} \approx 1 + \frac{1}{720} = 1.001\,388\,\bar{8}$$

$$\text{CODATA: } 1.001\,378\,419 \dots, \quad \Delta = +0.000\,010\,47 \text{ (}0.00105\%)$$

## 5. Kochenov Identity

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$$\boxed{\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \mathcal{O}(10^{-8})}$$

Numerically:

$$\epsilon_{\text{Koch}} = 144 - \frac{\phi^{12}}{\sqrt{5}} = -0.001\,388\,875\,493\dots$$

$$\delta_{\text{Koch}} = -\frac{1}{720} = -0.001\,388\,888\,889\dots$$

$$\epsilon_{\text{rem}} = \delta_{\text{Koch}} - \epsilon_{\text{Koch}} = -1.339\,56 \times 10^{-8}$$

$$|\epsilon_{\text{rem}}| \approx 1.34 \times 10^{-8}$$

Thus:

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} - 1.339\,56 \times 10^{-8}$$

## Notations

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|                                                     |                                                           |
|-----------------------------------------------------|-----------------------------------------------------------|
| $\phi = (1 + \sqrt{5})/2$                           | golden ratio                                              |
| $\sqrt{5} = 2.236\,067\,977\,5$                     |                                                           |
| $\sqrt{3} = 1.732\,050\,807\,57$                    |                                                           |
| $144 = F_{12}$                                      | twelfth Fibonacci number                                  |
| $720 = 6!$                                          | factorial of six; also $720 = \text{fermion spin return}$ |
| $31 = 2^5 - 1$                                      | Mersenne prime, discrete anchor                           |
| 100, 1000                                           | human scale (kilogram = litre of water)                   |
| $86400 = 120 \times 720$                            | seconds in a day                                          |
| $\epsilon_{\text{rem}} \approx 1.34 \times 10^{-8}$ | Kochenov Precision Boundary                               |

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See also (Zenodo, 2026): <https://zenodo.org/records/19769041> · <https://zenodo.org/records/20009102> · <https://zenodo.org/records/20009474>

# Kochenov Framework

Geometric Structure of Fundamental Constants

Collected Papers, 2026

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*$G, c, \alpha^{-1}, m_n/m_p, v_{\oplus}$  —  
all from  $\phi$ , 31, and  $1/720$ .*

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## Contents

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|          |                                                                                                        |          |
|----------|--------------------------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Platonic G: Gravitation from Geometry</b>                                                           | <b>2</b> |
| <b>2</b> | <b>The Integer 31 as a Discrete Geometric Node in Kochenov G</b>                                       | <b>3</b> |
| <b>3</b> | <b>The Gravitational Constant from Integers 31 and 144</b>                                             | <b>4</b> |
| <b>4</b> | <b>Geometric Origin of <math>10^{11}</math> in the Gravitational Constant</b>                          | <b>4</b> |
| <b>5</b> | <b>Geometric G and Its Formal Substitution into Einstein's Equations</b>                               | <b>5</b> |
| <b>6</b> | <b>Geometric G: From Earth's Geometry to the Gravitational Constant</b>                                | <b>6</b> |
| <b>7</b> | <b><math>m_n/m_p = 1 + 1/720</math>: A Simple Approximation</b>                                        | <b>7</b> |
| <b>8</b> | <b>A High-Precision Identity: <math>\phi^{12}/\sqrt{5} = 144 + 1/720 + \mathcal{O}(10^{-8})</math></b> | <b>8</b> |
| <b>9</b> | <b>Earth's Orbital Speed from Three Radii</b>                                                          | <b>8</b> |
|          | <b>Zenodo Records</b>                                                                                  | <b>9</b> |

## Platonic G: Gravitation from Geometry

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**Abstract.**  $G$  from geometry. No CODATA. Only golden ratio ( $\phi$ ), integer 31, and  $\sqrt{5}$ .

$$G_K = \frac{31^2 \times \sqrt{5}}{\phi^{12} \times 10^{11}} \approx 6.67354674496 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

### 1. Main Result

| Source              | $G \times 10^{-11}$   | Gap from $G_K$ |
|---------------------|-----------------------|----------------|
| Schlamminger (2014) | $6.6729 \pm 0.0005$   | +0.010%        |
| $G_K$ (this work)   | 6.67354674496         | —              |
| CODATA 2018         | $6.67430 \pm 0.00015$ | −0.011%        |

Labs disagree by 0.021%.  $G_K$  sits right in between. Inputs: only  $\phi$ , 31,  $\sqrt{5}$ . No fitting. No measured constants.

### 2. Generator Identity

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \mathcal{O}(10^{-8})$$

$720 = 6!$  — a weird intersection of three sequences (exponential  $\phi^n$ , factorial  $m!$ , Fibonacci  $F_n$ ) at  $n = 12$ ,  $m = 6$ .  $|\Delta| \approx 1.39 \times 10^{-8}$ , relative  $|\Delta|/144 \approx 9.7 \times 10^{-11}$ .

### 3. The 1/720 Residual — Three Domains

| Domain          | Formula                                                   |
|-----------------|-----------------------------------------------------------|
| Geometry        | $\phi^{12}/\sqrt{5} - 144 = 1/720 + \mathcal{O}(10^{-8})$ |
| Nuclear         | $m_n = m_p \times (1 + 1/720)$                            |
| Electromagnetic | $\alpha^{-1} = 144 - 7 - 1/720 \approx 137$               |

### 4. Open Question

$31 = 2^5 - 1$  (Mersenne prime). Why 31? Nobody knows yet.

### 5. Full Precision Reference

$$G_K = 6.6735467449616181237591997717904236 \dots \times 10^{-11}$$

## The Integer 31 as a Discrete Geometric Node in Kochenov G

**Abstract.** The integer 31 is the best integer for two geometric rules: (1) Quadratic closure:  $31^2 = G_K \times 10^{11} \times 144$ , gap 0.001%. (2) Nearest integer to the continuous best  $n \approx 31.338$  from  $(\phi^6 \times n)^4 = 10^{11}$ . Neighbours 30 and 32 fail both.

### 1. Two Geometric Rules for 31

**Rule A — Quadratic closure:**

$$31^2 = 961 = G_K \times 10^{11} \times 144, \quad \text{gap } 0.001\%$$

**Rule B — Scaling optimum:**

$$(\phi^6 \times n)^4 = 10^{11} \Rightarrow n \approx 31.338, \quad \text{nearest integer} = 31$$

### 2. Comparison with Neighbours

| $n$       | $G$ gap from CODATA | $(\phi^6 n)^4$ gap from $10^{11}$ | Note                      |
|-----------|---------------------|-----------------------------------|---------------------------|
| 30        | 6.358%              | 16.018%                           | too small                 |
| <b>31</b> | <b>0.011%</b>       | <b>4.247%</b>                     | <b>best discrete node</b> |
| 32        | 6.544%              | 8.718%                            | too large                 |
| 31.338    | $\sim 0\%$          | $\sim 0\%$                        | continuous optimum        |

### 3. Extra Connection

$$31^2 = 961 \approx 367 \times \phi^2 \quad (\text{gap } 0.019\%)$$

This allows writing  $G$  without an explicit  $31^2$ :

$$G_K = \frac{367 \sqrt{5}}{\phi^{10} \times 10^{11}}, \quad \text{gap } 0.019\%$$

Also:  $31 = (3 \times L_{12} - 5)$  where  $L_{12} = 322$  is the 12th Lucas number (twin of  $F_{12} = 144$ ). So 31 is not a guess.

### 4. Full Precision Reference

$$\begin{aligned}
 G_K &= \frac{31^2 \times \sqrt{5}}{\phi^{12} \times 10^{11}} \\
 &= 6.6735467449616181237591997717904236079132174548715155655761367183 \\
 &\quad \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}
 \end{aligned}$$

## The Gravitational Constant from Integers 31 and 144

**Abstract.**  $G$  from two integers: 31 and 144. Core:  $G_{\text{core}} = (31^2/144) \times 10^{-11} \approx 6.673611 \times 10^{-11}$ . Refined:  $G_K = (31^2/(144 + 1/720)) \times 10^{-11} \approx 6.67354674496 \times 10^{-11}$ . No CODATA fitting. Just integers and  $1/720$ .

### Integer Core

$$G_{\text{core}} = \frac{31^2}{144} \times 10^{-11} = \frac{961}{144} \times 10^{-11} \approx 6.673611 \times 10^{-11}$$

### Refined Discrete Form

$$G_K = \frac{31^2}{144 + 1/720} \times 10^{-11} \approx 6.67354674496 \times 10^{-11}$$

### Comparison with Reference Values

| Source                               | $G \times 10^{-11}$ | Gap from $G_K$ |
|--------------------------------------|---------------------|----------------|
| Huazhong 2007                        | 6.67349             | +0.00085%      |
| $G_K$ (this work)                    | 6.67354674496       | —              |
| $G_{\text{core}}$ (this work)        | 6.673611111         | −0.00096%      |
| NIST 2026 (Schlamminger et al.)      | 6.67387             | −0.00484%      |
| CODATA 2018                          | 6.67430             | −0.0113%       |
| NIST 2026 vs Huazhong 2007 differ by |                     | 0.0235%        |

$G_K$  and  $G_{\text{core}}$  both sit inside the range of lab measurements, from pure math, not from fitting.

### Relation to the Golden Ratio

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \epsilon, \quad |\epsilon| < 10^{-8}$$

gives the geometric bridge:  $G_K = (31^2 \times \sqrt{5})/(\phi^{12} \times 10^{11})$ .

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## Geometric Origin of $10^{11}$ in the Gravitational Constant

**Abstract.** The factor  $10^{11}$  in  $G$  is not a decimal accident. It comes from four turns of a 3D golden spiral plus a breathing step of  $1/720$ . Match: 0.020%.

## 1. The Spiral Block

$$S = \phi^6 \times 31 \approx 556.272\,429, \quad \kappa = \log_{10} S \approx 2.745\,288$$

$$4\kappa \approx 10.981\,150, \quad \log_{10}(10^{11}) = 11, \quad \text{gap} = 0.018\,850$$

## 2. Micro-step (Breathing Term)

One turn has 12 micro-steps, each =  $1/720$ :

$$12 \times \frac{1}{720} = \frac{1}{60} \approx 0.016\,667$$

$$10.981\,150 + 0.016\,667 = 10.997\,817, \quad \text{residual} = 0.002\,183 \approx 0.020\%$$

## 3. Resonant Relation

$$\log_{10}(10^{11}) \approx 4 \times \log_{10}(\phi^6 \times 31) + \frac{12}{720}, \quad \text{holds to } 0.020\%$$

## 4. Four Turns of the 3D Spiral

The leftover  $\Delta_{\text{fractal}} = 11 - (4\kappa + 12/720) \approx 0.002183$  is not a calculation error. It is the seam where binary meets golden spiral.

Binary shift:  $2^5 = 32$  acts as outer locking ring, forcing the anchor to 31. The squeeze from 32 to 31 gives exactly this log leftover.

Connections:

$$\Delta_{\text{fractal}} \times 144 \approx \frac{\pi}{10}, \quad \Delta_{\text{fractal}} \times 720 \approx \frac{\pi}{2}, \quad \Delta_{\text{fractal}} \approx \frac{\pi}{1440}$$

## 5. Equivalent Forms of $G_K$

$$G_K = \frac{31^2 \times \sqrt{5}}{\phi^{12} \times 10^{11}} = \frac{31^2}{(144 + 1/720) \times 10^{11}} \approx 6.67354674496 \times 10^{-11}$$

# Geometric G and Its Formal Substitution into Einstein's Equations

**Abstract.**  $G$  from  $\phi$ , 31, and  $\sqrt{5}$ . In Earth-based units (meter from meridian, kilogram from liter of water),  $G$  becomes pure  $\text{s}^{-2}$ :

$$G_{\text{Earth}} = \frac{31^2 \times \sqrt{5}}{\phi^{12}} \times 10^{-8} \text{ s}^{-2}$$

Plug into Einstein's equations. Just substitution. No change to GR.



## 1. Formal Substitution

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \longrightarrow G_{\mu\nu} = \frac{8\pi \times 31^2 \times \sqrt{5}}{\phi^{12} \times 10^{11}} T_{\mu\nu}$$

Same equations. Different look.

## 2. Earth-Based Units: Killing Kilograms and Meters

$$1 \text{ m} = M/10^7, \quad 1 \text{ kg} \approx M^3/10^{24}, \quad \frac{1 \text{ m}^3}{1 \text{ kg}} \approx 10^3$$

So  $G_{\text{Earth}} = G_K \times 10^3 = (31^2 \times \sqrt{5})/\phi^{12} \times 10^{-8} \text{ s}^{-2}$ . All meters and kilograms gone.

Note:  $86400 \text{ s/day} = 120 \times 720 = 120 \times 6!$  The same 720 appears in  $\phi^{12}/\sqrt{5} = 144 + 1/720$ .

## 3. Two Rules that Pick 31

**Rule 1:**  $31^2 = G_K \times 10^{11} \times 144$ , gap 0.001%.

**Rule 2:**  $(\phi^6 \times n)^4 = 10^{11}$  gives  $n = 31.338$ ; nearest integer = 31.

## Geometric G: From Earth's Geometry to the Gravitational Constant

**Abstract.** In units from Earth (meter from meridian, kilogram from water, second from day), the meters and kilograms cancel. You're left with just  $1/\text{s}^2$ :

$$G_{\text{Earth}} = \frac{31^2 \times \sqrt{5}}{\phi^{12}} \times 10^{-8} \text{ s}^{-2} \approx 6.67354674496 \times 10^{-8} \text{ s}^{-2}$$

In SI, this sits right between Schlamminger (2014) and CODATA 2018. No experiments used. Just geometry.

## Connection of 720 to Earth's Rotation

$$86400 \text{ s/day} = 120 \times 720, \quad \frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \epsilon, \quad |\epsilon| \approx 1.39 \times 10^{-8}$$

The same 720 in Earth's rotation and in the golden ratio identity.

## Comparison

| Source                         | $G \times 10^{-11}$   | Gap from $G_{\text{Earth}}$ |
|--------------------------------|-----------------------|-----------------------------|
| Schlamminger (2014)            | $6.6729 \pm 0.0005$   | +0.010%                     |
| $G_{\text{Earth}}$ (this work) | 6.67354674496         | —                           |
| CODATA 2018                    | $6.67430 \pm 0.00015$ | −0.011%                     |
| Labs disagree by               |                       | 0.021%                      |

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## $m_n/m_p = 1 + 1/720$ : A Simple Approximation

**Abstract.** The neutron-proton mass ratio is approximately  $1 + 1/720$ . The same  $1/720$  appears in  $G_K$ , the golden-ratio identity, and the fine-structure constant. Why? Nobody knows yet.

### 1. Main Result

$$\frac{m_n}{m_p} \approx 1 + \frac{1}{720} = 1.001388\bar{8}$$

CODATA:  $m_n/m_p = 1.00137841946 \dots$  Difference:  $\approx 0.00105\%$ .

### 2. Context: $1/720$ in Four Domains

- $G_K$  (refined gravitational constant)
- Golden-ratio identity:  $\phi^{12}/\sqrt{5} = 144 + 1/720 + \epsilon$ ,  $|\epsilon| < 10^{-8}$
- Fine-structure constant:  $\alpha^{-1} = 144 - 7 - 1/720 \approx 137$
- Neutron-proton ratio:  $m_n/m_p = 1 + 1/720$

Pure math, gravity, nuclear physics. Same factorial 720. Weird.

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## A High-Precision Identity: $\phi^{12}/\sqrt{5} = 144 + 1/720 + \mathcal{O}(10^{-8})$

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**Abstract.**  $\phi^{12}/\sqrt{5}$  is almost exactly  $144 + 1/720$ . The difference is  $\sim 10^{-8}$ .

### 1. The Numbers

$$X = \frac{\phi^{12}}{\sqrt{5}} \approx 144.001388875432\dots$$

$$144 + \frac{1}{720} = 144.001388\bar{8}$$

$$\Delta = X - \left(144 + \frac{1}{720}\right) = -1.394 \times 10^{-8}, \quad |\Delta|/X \approx 9.68 \times 10^{-11}$$

### 2. Where 144 Comes From (Binet's Formula)

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}, \quad \psi = -1/\phi$$

For  $n = 12$ :  $F_{12} = 144$ . Rearranging:

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{\phi^{-12}}{\sqrt{5}}, \quad \frac{\phi^{-12}}{\sqrt{5}} \approx 0.001388875 \approx \frac{1}{720}$$

### 3. The Kochenov Epsilon

$$\phi^{12}/\sqrt{5} = 144 + \frac{1}{720} + \epsilon, \quad \epsilon \approx -1.34 \times 10^{-8}$$

This is not a flaw. It is the “diversity term” — the irreducible seam that keeps the universe from being a dead crystal.

$$\text{Perfect geometry: } 144 + \frac{1}{720} \quad \text{Real universe: } 144 + \frac{1}{720} + \epsilon$$


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## Earth's Orbital Speed from Three Radii

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**Abstract.** A geometric relation between solar system radii:

$$\frac{R_{\odot} \times R_{\text{Moon}}}{R_{\oplus}^2} \approx 29.78 \text{ km/s}$$

Earth's orbital speed is 29.78 km/s. Difference: 0.013%. No masses. No  $G$ . No Kepler. Just lengths.

**Numbers (IAU 2026 mean radii)**

$$R_{\odot} = 695,700.0 \text{ km}, \quad R_{\text{Moon}} = 1,737.4 \text{ km}, \quad R_{\oplus} = 6,371.0 \text{ km}$$

$$\frac{R_{\odot} \times R_{\text{Moon}}}{R_{\oplus}^2} = \frac{1.20870918 \times 10^9}{4.05896410 \times 10^7} \approx 29.7788 \text{ km/s}, \quad v_{\oplus} = 29.7827 \text{ km/s}, \quad \Delta = 0.013\%$$

**Why the Units Don't Kill This**

In a consistent Earth-based unit system (length from meridian, time from equatorial rotation), velocity becomes dimensionless. The relation becomes a pure geometric statement about three radii.

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**Zenodo Records (April–May 2026)**

- $\phi^{12}/\sqrt{5} = 144 + 1/720$ : <https://zenodo.org/records/19769041>
- Kochenov Geometric Identity for  $G$ : <https://zenodo.org/records/20009102>
- Geometric Origin of  $10^{11}$  in  $G$ : <https://zenodo.org/records/20009474>
- The Integer 31 as a Discrete Node: <https://zenodo.org/records/20071070>
- $G$  from Integers 31 and 144: <https://doi.org/10.5281/zenodo.20117611>
- $m_n/m_p = 1 + 1/720$ : <https://doi.org/10.5281/zenodo.20146556>

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